Campus Road Safety Observatory: An AI-based Platform for Hazard Reporting and Safety Analytics

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1. Problem Statement

Road safety issues on university campuses such as potholes, damaged footpaths, faded road markings, poor lighting, drainage problems, unsafe intersections, and speeding are often identified only after they become serious hazards. IIT Mandi currently has no centralized digital platform that enables students, faculty, and staff to systematically report, visualize, and prioritize these issues.

This project aims to develop a prototype Campus Road Safety Observatory that combines community-based reporting, AI-assisted image classification, and an interactive GIS dashboard. The platform is intended to help campus authorities identify hazard hotspots, prioritize maintenance activities, and support a safer campus environment.

2. Project Objectives

- Develop an integrated digital platform (mobile-responsive web app + centralized database) for crowdsourced reporting of campus road safety hazards, including photo upload and automatic GPS geotagging.
- Develop an AI-based image classification module to automatically categorize reported hazards (e.g. potholes, damaged pavements, faded markings, drainage issues).
- Develop an interactive GIS dashboard to visualize, analyze, and prioritize reported hazards, supporting campus authorities in infrastructure planning and maintenance.

3. Project Scope – What Will the Students Do?

The student team will design, develop, and demonstrate a functional prototype comprising four core components.

3.1 Hazard Reporting Application

A mobile-responsive web application that allows users to report hazards, upload photographs, automatically capture GPS location, select hazard categories, and track the status of reported issues.

3.2 AI-assisted Image Classification

A basic image classification module, built using transfer learning on existing deep learning models, that automatically categorizes uploaded images into classes such as:

- Potholes

- Damaged pavements
- Faded road markings
- Roadside obstructions
- Drainage issues
- Other hazards

3.3 Interactive Safety Dashboard

A dashboard for campus authorities that includes a GIS-based campus map, heat maps showing hazard hotspots, filters based on hazard category and date, summary statistics and trends, and priority ranking of reported issues.

3.4 Database and Analytics

A backend database to store reported hazards, manage user reports, generate summary reports, and support future expansion of the platform.

4. Team Structure & Role Distribution

The project splits naturally into four parallel sub-teams. Depending on team size, a student may contribute to more than one area, but each area should have a clear owner.

Table 1. Responsibilities and Skillset Needed

Sub-team	Core responsibilities	Key skills / tools
Web / App Development	Mobile-responsive reporting application: photo upload, GPS capture, category selection, status tracking	React/Flutter, REST APIs, geolocation APIs
AI / Computer Vision	Image classification module for hazard categories using transfer learning; ongoing model refinement	Python, PyTorch/TensorFlow, pretrained CNNs (ResNet, YOLOv8)
GIS & Dashboard	Interactive map, heat maps, filters, summary analytics, priority ranking for campus authorities	Leaflet/Mapbox, GeoJSON, charting libraries
Database & Backend	Schema design, storage of reports/images, APIs connecting frontend, CV module, and dashboard	PostgreSQL/MongoDB, Node.js/Django backend

5. Semester Timeline & Milestones

Based on a 14-week semester. The following plan is developed:

Table 2. Work Plan for App Development

Week	Activities
1	Team formation, role assignment, tool & literature survey (existing hazard apps, YOLOv8/ResNet, Leaflet/Mapbox)

2	Finalize hazard taxonomy and data schema (report fields, categories, priority levels)
3-6	Begin campus data collection: geotagged hazard photos along assigned routes
4-5	Build reporting web app skeleton (photo upload, GPS capture, category selection)
5-6	Set up backend database
6-7	Begin computer vision module: transfer learning on collected images + supplementary public datasets
8	Integrate reporting app with database; working end-to-end submission pipeline
9	Build GIS dashboard skeleton: campus map with hazard markers
10	Add heat maps, filters (category/date), summary statistics
11-12	Integrate computer vision output into pipeline (auto-tagging); add priority ranking logic
13-14	Field test on selected campus roads; refine accuracy using misclassified samples

6. Expected Deliverables

The following deliverables are expected from this project:

- Functional hazard reporting application
- AI-assisted image classification module
- Interactive GIS-based dashboard with heat maps
- Database of reported safety issues
- Prototype demonstration on selected IIT Mandi roads
- Technical documentation and source code

7. Expected Learning Outcomes

Students will gain hands-on experience in:

- Application development
- Application of Artificial Intelligence and Computer Vision
- GIS and spatial data visualization
- Database management
- Human-centered design
- Transportation engineering and road safety
- Team-based product development